

Project 3: Active Learning for Learning-to-Defer

Task 1: Baseline AI – 91.21%

For the first task, I used the AG News dataset, which contains four classes: World, Sports, Business and Sci/Tech. The goal was to train a model that can predict the category of a news article.

I converted the text of the articles into numerical features using TF-IDF. I used a maximum of 10,000 features and removed common English stop words. After converting the text, I trained a Logistic Regression model on the full training dataset.

The model was evaluated on the test set and achieved an accuracy of **91.21%**. I used this result as the baseline for comparing the other methods in the following tasks.

Task 2: Simulated Expert – 72.14%

For the second task, I created a simulated expert. The expert was intentionally designed to be better at some categories than others instead of being equally accurate for every type of article.

The expert was configured with the following probability of giving the correct answer:

- Sports: **95%**
- Business: **90%**
- World: **60%**
- Sci/Tech: **40%**

This means that the expert is stronger for Sports and Business and weaker for World and Sci/Tech. I designed it this way because the expert should have specific areas of expertise and should not be perfect.

The simulated expert achieved an overall test accuracy of **72.14%**. Although this was lower than the baseline AI, the expert can still be useful for some examples where the AI makes an incorrect prediction.

Task 3: Learning-to-Defer – 88.22%

For Task 3, I created a learning-to-defer system where the system decides whether the AI should make the prediction itself or whether the article should be given to the simulated expert.

First, I generated predictions from both the baseline AI and the simulated expert on the training data. An example was marked as useful for deferral when the baseline AI predicted the wrong class but the expert predicted the correct class.

I then trained another Logistic Regression model as an allocator. The purpose of the allocator is to learn from the article features when it is better to use the AI and when it is better to defer to the expert.

During testing, if the allocator selected deferral, the prediction from the simulated expert was used. Otherwise, the prediction from the baseline AI was used.

The final Human-AI team achieved an accuracy of **88.22%**.

This result was lower than the baseline AI accuracy of 91.21%. One reason is that the simulated expert has a lower overall accuracy, so some articles that were deferred to the expert were still classified incorrectly. The program also records how many test articles were deferred, which helps to understand the behaviour of the deferral system.

Task 4: Active Learning – 86.80%

For Task 4, I used an active learning method based on **uncertainty sampling**.

In this task, the system does not have access to expert predictions for the whole training dataset. Instead, I used a limited budget of **2,000 expert queries**.

First, I calculated the class probabilities produced by the baseline AI for every training article. I measured the uncertainty of each prediction using:

Uncertainty = 1 – probability of the AI's most confident prediction

A high uncertainty value means that the AI is not very confident about its prediction.

I then sorted the articles from the most uncertain to the least uncertain and selected the top 2,000 articles. The simulated expert was queried only for these selected examples.

Using these limited expert predictions, I created the deferral labels in the same way as Task 3. An example was considered useful for deferral when the AI was wrong and the expert was correct. I then trained a new Logistic Regression allocator using only these queried examples.

The Active Learning Human-AI team achieved an accuracy of **86.80%**.

I chose uncertainty sampling because the articles where the AI has low confidence are likely to be useful examples for asking the expert. This also reduces the number of expert queries compared with having expert labels for the full training dataset.

Final Results

Method	Test Accuracy
Baseline AI	91.21%
Simulated Expert	72.14%
Learning-to-Defer Team	88.22%
Active Learning Team	86.80%

The baseline AI achieved the highest overall accuracy. The simulated expert had lower overall performance, but it was designed to have specific strengths and weaknesses. The learning-to-defer system attempted to combine the AI and the expert by learning when the expert could be useful.

The active learning system used a limited number of expert queries and focused on the examples where the AI was most uncertain. Although the Human-AI and Active Learning results were lower than the baseline in this experiment, the project shows how an AI system can learn when to use an expert and how active learning can reduce the amount of expert information required.